A nontrivial degree upper bound for square-root polynomials for every odd prime

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Allowed standard prerequisites (editorial).

- Polynomial vanishing sets over finite fields and formal differentiation. Author Sections 1-3; original full bibliography retained
- Weil bound for mixed character sums and Fourier L1 bounds for intervals. Author Section 5.1 and original Schmidtbook reference; local equidistribution proof included

Theorem 5.1. *For all odd primes p, for $t = o\left(\frac{\sqrt{p}}{\log^2 p}\right)$, there is a polynomial of degree $\frac{p}{2} - t$ which computes square roots.*

Let p be an odd prime, and let $\mathbb{F}_p$ be the finite field with p elements.

We say $f(X) \in \mathbb{F}_p[X]$ *computes square roots* in $\mathbb{F}_p$, if for all nonzero perfect squares $a \in \mathbb{F}_p$, we have:

$$f(a)^2 = a.$$

In other words, for each nonzero perfect square $a \in \mathbb{F}_p$, $f(a)$ is one of its two square roots.

6 Upper bounds for general p

In this section, we give a slightly nontrivial upper bound on the degree of polynomials computing square roots for all p. We will show that there is a polynomial with degree somewhat less than $\frac{p}{2}$ which computes square roots.

Proof. Let $m = (p-1)/2$. Let $S \subseteq \mathbb{F}_p$ be the set of nonzero perfect squares, and note that $|S| = m$. For each $\alpha \in S$, let $\delta_\alpha(X) \in \mathbb{F}_p[X]$ be the unique polynomial of degree $\leq (m-1)$ such that for all $\beta \in S$:

$$\delta_\alpha(\beta) = \begin{cases} 1 & \beta = \alpha, \\ 0 & \beta \neq \alpha. \end{cases}$$

Explicitly, we have:

$$\delta_\alpha(X) = \frac{1}{m}\left(\left(\frac{X}{\alpha}\right)^{m-1} + \left(\frac{X}{\alpha}\right)^{m-2} + \ldots + \frac{X}{\alpha} + 1\right).$$

Then given a function $u : S \to \mathbb{F}_p$, the unique polynomial $f(X) \in \mathbb{F}_p[X]$ of degree at most $m-1$ such that $f(\alpha) = u(\alpha)$ for all $\alpha \in S$ is given by:

$$f(X) = \sum_{\alpha \in S} u(\alpha)\delta_\alpha(X).$$

Our goal is to pick u where each $u(\alpha)$ is one of the two square roots of α so that many of the leading coefficients of $f(X)$ equal 0.

We now use the structure of S. Let g be a generator of $\mathbb{F}_p^*$. Then $S = \{g^{2j} \mid 0 \le j < (p-1)/2\}$. Furthermore, for $\alpha = g^{2j} \in S$, one of the two square roots of α is g^j.

Thus, our problem can be reformulated as choosing a function $v : S \to \{\pm 1\}$ such that:

$$f(X) = \sum_{j=0}^{(p-1)/2} v\left(g^{2j}\right) \cdot g^j \cdot \delta_{g^{2j}}(X)$$

has many leading coefficients equal to 0.

Observe that the coefficient of X^{m-i} in $f(X)$ equals:

$$\frac{1}{m} \sum_{j=0}^{(p-1)/2} v(g^{2j}) g^j \left(\frac{1}{g^{2j}}\right)^{m-i} = \frac{1}{m} \sum_{i=0}^{(p-1)/2} v(g^{2j}) g^{(2i+1)j}.$$

Thus, to get a polynomial $f(X)$ of degree $< m-t$, we want to find a vector $v \in \{\pm 1\}^m$ that lies in the kernel of the Vandermonde-type matrix $M \in \mathbb{F}_p^{t \times m}$, where:

$$M_{i,j} = g^{(2i+1)j}.$$

(The row index i runs from 1 to t, the column index j runs from 0 to $m-1$.)

For later use, for a vector $y \in \mathbb{F}_p^t$, we define $P_y(Z) \in \mathbb{F}_p[Z]$ to be the polynomial:

$$P_y(Z) = \sum_{i=1}^{t} y_i Z^{2i+1}.$$

Thus for $j \in \{0, 1, \ldots, m-1\}$, the jth entry of $M^T y$ equals $P_y(g^j)$.

To show that there exists the desired ± 1 vector, we count the number of such vectors using Fourier analysis. Let ω be a pth root of unity in $\mathbb{C}$. The

number of such ± 1 vectors equals:

$$\begin{aligned}
N &= \sum_{v \in \{\pm 1\}^m} \mathbf{1}_{Mv=0} \\
&= \sum_{v \in \{\pm 1\}^m} \mathbb{E}_{y \in \mathbb{F}_p^t} \left[\omega^{\langle y, Mv \rangle} \right] \\
&= \mathbb{E}_y \left[\sum_v \omega^{\langle M^T y, v \rangle} \right] \\
&= \mathbb{E}_y \left[\sum_v \omega^{\sum_{j=0}^{m-1} (M^T y)_j \cdot v_j} \right] \\
&= \mathbb{E}_y \left[\sum_v \prod_{j=0}^{m-1} \omega^{(M^T y)_j \cdot v_j} \right] \\
&= \mathbb{E}_y \left[\sum_v \prod_{j=0}^{m-1} \omega^{P_y(g^j) \cdot v_j} \right].
\end{aligned}$$

For $y = 0$, the expression inside the expectation equals 2^m. We will show that for the remaining $p^t - 1$ values of y, the expression inside the expectation is very small.

Fix any $y \neq 0$. The expression inside the expectation equals:

$$\sum_{v \in \{\pm 1\}^m} \prod_{j=0}^{m-1} \omega^{P_y(g^j) \cdot v_j} = \prod_{j=1}^{m} \left(\omega^{P_y(g^j)} + \omega^{-P_y(g^j)} \right). \tag{1}$$

The next lemma (which uses the Weil bounds on mixed character sums) shows that for any nonzero y, the evaluations of the polynomial P_y at $\{1, g, g^2, \ldots, g^{m-1}\}$ are well distributed in $\mathbb{F}_p$.

Lemma 6.1. *Let $0 \leq \alpha < \beta \leq 1$. Let $y \in \mathbb{F}_p^t \setminus \{0\}$. Then:*

$$\Pr_{j \in \{0,1,\ldots,m-1\}} \left[P_y(g^j) \in [\alpha p, \beta p] \right] = (\beta - \alpha) + O\left(\frac{t \log^2 p}{\sqrt{p}} \right).$$

Assuming the lemma, we get that for $t = o\left(\frac{\sqrt{p}}{\log^2 p} \right)$, the product in Equation (1) is at most $2^m \cdot \exp(-m)$. Thus:

$$\begin{aligned}
N &\geq \frac{2^m}{p^t} - \max_{y \neq 0} \left| \prod_{j=1}^{m} \left(\omega^{P_y(g^j)} + \omega^{-P_y(g^j)} \right) \right| \\
&\geq \frac{2^m}{p^t} - O(2^m \exp(-m)) \\
&\geq 2^m \left(\frac{1}{p^t} - \exp(-m) \right) \\
&> 0,
\end{aligned}$$

where the penultimate inequality holds since

$$t = o\left(\frac{\sqrt{p}}{\log^2 p}\right) \ll \frac{p}{\log p} = \Theta\left(\frac{m}{\log p}\right).$$

This completes the proof. □

6.1 Distribution of values of $P_y(g^j)$

We now prove Lemma 6.1.

Proof. Let I be the interval $[\alpha p, \beta p]$. Let J be the set $\{1, g, g^2, \ldots, g^{m-1}\}$. Then the probability in the statement of the lemma equals:

$$\frac{1}{m} \sum_{z \in \mathbb{F}_p} \mathbf{1}_I(P_y(z)) \mathbf{1}_J(z). \tag{2}$$

I is an interval in the additive group of $\mathbb{F}_p$ By standard results, if we expand $\mathbf{1}_I$ in its additive Fourier expansion:

$$\mathbf{1}_I = \sum_{\psi} \widehat{\mathbf{1}}_I(\psi)\psi$$

(where the ψ are the additive characters), then:

$$\sum_{\psi} |\widehat{\mathbf{1}}_I(\psi)| \leq O(\log p). \tag{3}$$

Similarly, J is an interval in the multiplicative group of $\mathbb{F}_p$. If we expand $\mathbf{1}_J$ in its additive Fourier expansion:

$$\mathbf{1}_J = \sum_{\psi} \widetilde{\mathbf{1}}_J(\chi)\chi$$

(where the χ are the multiplicative characters), then:

$$\sum_{\chi} |\widetilde{\mathbf{1}}_I(\chi)| \leq O(\log p). \tag{4}$$

Then the probability in Equation (2) equals:

$$\begin{aligned}
&\frac{1}{m} \sum_z \left(\sum_{\psi} \widehat{\mathbf{1}}_I(\psi)\psi(P_y(z)) \right) \left(\sum_{\chi} \widetilde{\mathbf{1}}_J(\chi)\chi(z) \right) \\
&= \frac{1}{m} \sum_{\psi,\chi} \widehat{\mathbf{1}}_I(\psi)\widetilde{\mathbf{1}}_J(\chi) \left(\sum_z \psi(P_y(z))\chi(z) \right) \\
&= \frac{1}{m} \left(\frac{|I|}{p} \cdot \frac{|J|}{p} \cdot p + O\left(\sum_{(\psi,\chi)\neq(1,1)} |\widehat{\mathbf{1}}_I(\psi)| \cdot |\widetilde{\mathbf{1}}_J(\chi)| \cdot \left| \sum_z \psi(P_y(z))\chi(z) \right| \right) \right) \\
&= (\beta - \alpha) + \frac{1}{m} \cdot O\left(\sum_{(\psi,\chi)\neq(1,1)} |\widehat{\mathbf{1}}_I(\psi)| \cdot |\widetilde{\mathbf{1}}_J(\chi)| \cdot \left| \sum_z \psi(P_y(z))\chi(z) \right| \right)
\end{aligned}$$

The Weil bound for mixed character sums (see [Sch06]) shows that whenever (ψ, χ) are not both trivial characters, the inner expression is bounded:

$$\left| \sum_z \psi(P_y(z))\chi(z) \right| \leq O(t\sqrt{p}).$$

Combined with the bounds in Equations (3), (4), we get the desired bound on the probability. □

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